KISI-KISI SOAL / DESKRIPSI TEKNIS LOMBA KOMPETENSI SISWA (LKS) DIKMEN (SMK/SMA/MA/MAK) TINGKAT PROVINSI JAWA TIMUR TAHUN 2025

BIDANG LOMBA :

IT Software Solutions for Business

LKS DIKMEN
TINGKAT PROVINSI JAWA TIMUR
(Jenjang SMK/SMA/MA/MAK)

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Kisi-Kisi LKS Provinsi 2025:

IT Software Solutions for Business

A. Hardware Recommendation

Tools	Specification
Desktop/Laptop	• Processor 2.0 GHz Quad core or higher • RAM 16 GB or higher • HDD/SSD 256 GB or higher • Operating System Windows 10
Mouse	Wired Mouse Optic (No memory or Macro capability allowed)
Keyboard	Wired Keyboard (No memory or Macro capability allowed)
Monitor	20 - 24 inch
Dongle Wi-Fi	For PC User

B. Required Software

Software	Specification
Windows 10, 64 bits	Windows 10 Education 64 Bit
Microsoft Office 365 Home/Education	*Free Trial* 1 *month*
Visual Studio Community Edition 2022 • ASP .NET & Web Development ✓ .NET Framework 4.7.X or higher ✓ .NET 6.0 or .NET Core 6.0 • .NET Desktop Development ✓ .NET Framework 4.7.X or higher	-
Microsoft SQL Server Express Edition 2019	-
SQL Server Management Studio 19.x	-
Android Studio Ladybug • Android SDK Tools • Android SDK Platform-Tools • Android SDK Build-Tools • Android SDK Platform • Kotlin • AVD Manager (PIXEL XL API 33) • Android SDK API 33	-
Google Chrome Browser (latest version)	-

C. General Requirements:

- **Test project will be given in English. Dictionary (hardcopy) is allowed during the competition.**
- Every module should be provided with proper validation and informative error messages.
- Scrollbar should be visible whenever there are lists or views that occupy more than one screen.
- Use proper naming conventions for variables, functions, or filename.
- By default, form should be displayed in the centre of the screen.
- Whenever a form or dialog is in focus, other form functionality should be disabled.
- Wireframe given in the test project is for reference only. Modifications are allowed if they do not affect the performance and functionality of the requested features.
- Pay attention to given working durations. The time given includes the duration for the submission process. Additional time will not be added during normal conditions.

D. Test Project Detail

Module Name	Duration (Hours)	Requirements
Desktop Application (40%)	**3**	• Import database and its content from external sources. • Create windows forms and its components based on given requirements. • Validating user input. • Role validation and menu access restriction based on login data. • Show, insert, update, or delete records from the database. • Create windows forms and its components based on given requirements. • Join tables query. • Aggregate query. • Display data using charts.

		• Searching and filtering. • Datagridview manipulation.
Mobile Application (40%)	**3**	• Create a mobile application form and its component based on given requirements. • Show, insert, update, or delete records from external sources (database or API). • Searching and filtering. • Export data to external files. • Show Image from others resource. • Animation using canvas. • User Authentication and Token Management
Web API (20%)	**3**	• Analyze given requirements and translate them into an effective database design. • Develop RESTful API endpoints according to the provided specifications. • Implement the appropriate logic for each endpoint. • Ensure each RESTful API endpoint includes correct headers, parameters, request body, method, and suitable return objects (HTTP status code, response headers, and response body). • Handle error scenarios by providing meaningful and consumable responses for all potential negative cases. • Implement search and filtering functionality. • Manage user authentication and token handling securely.

E. Proposed Schedule

Day	Activities	Participant
C-1	Opening	• Juri • Peserta • Pembimbing
	Technical Meeting: 1. Competition Rules and Procedure 2. Technical Description Discussion 3. Test Project Briefing	• Juri • Teknisi • Peserta • Pembimbing
	Familiarization: 1. Workshop Setup and Checking 2. Hardware and Software Testing	• Juri • Teknisi • Peserta • Pembimbing (opsional)
C1	Competition Day 1 Session 1: Mobile Application	• Juri • Teknisi • Peserta
	Competition Day 1 Session 2: Desktop Application	
C2	Competition Day 2 Session 1: Web API Application	• Juri • Teknisi • Peserta
	Competition Day 2 Session 2: -	